

When Demographic Parity “Improves” Under Shift: A Simulation Study of Fairness Metric Distortion

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1 Problem Sketch

Many popular group fairness metrics are defined under a fixed data distribution. However, in real applications the distribution often drifts: group proportions change, feature distributions shift, and label noise may become asymmetric. In such settings, it is unclear how to interpret changes in fairness metrics: an apparent improvement in demographic parity might be an artifact of shift rather than a genuinely fairer model, while error-based metrics such as equalized odds may degrade sharply.

In this project, I systematically explore how different types of controlled distribution shift affect the behavior of standard group fairness metrics, even when the underlying classifier is kept fixed.

2 Research Questions

- **RQ1:** Under which types of distribution shift (prior / covariate / label noise) does Demographic Parity move toward zero while Equalized Odds worsens?
- **RQ2:** How does the severity of shift affect the magnitude of this discrepancy? Is there a threshold beyond which both metrics degrade together?
- **RQ3:** Can a simple post-hoc reweighting heuristic (e.g., importance weighting) reliably detect or correct the “mirage improvement”? Specifically, does it make DP’s movement better aligned with EO’s movement, or does it fail in some shift regimes?

3 Plan / Methodology

3.1 Base Setting

- Binary classification with two demographic groups A and B .
- Fixed simple classifier (logistic regression) trained on an initial (unshifted) distribution.
- **Main experiment:** Train once, evaluate on test sets with increasing shift severity.
- **Supplementary experiment (Appendix):** For each shift severity level, retrain the classifier on that shifted distribution to examine how much distortion is due to distribution mismatch vs. inherent classifier limits.

3.2 Shift Scenarios

1. **Group prior shift:** progressively change base rates / group proportions while keeping conditional feature distributions fixed.
2. **Covariate shift:** widen or translate the feature distribution of one group (e.g., increase variance or shift the mean along a relevant dimension).
3. **Asymmetric label noise:** introduce different label-flip probabilities for the two groups.

3.3 Metrics Tracked

- Demographic parity (difference in positive prediction rates).
- Equalized odds (TPR/FPR gaps).
- Calibration error per group (ECE gap between groups).

3.4 Heuristic

For label noise and covariate shift, we implement a simple **kernel density ratio estimator**

$$w(x) = \frac{\hat{p}_{\text{test}}(x)}{\hat{p}_{\text{train}}(x)}$$

and apply it to reweight the loss during evaluation (or to adjust predicted probabilities). We compare fairness metrics with and without this correction.

3.5 Severity Parameter Definitions

Table 1: Shift scenarios and their severity parameters.

Shift scenario	Severity parameter	Definition
Group prior shift	$\alpha \in [0, 1]$	$P(Y = 1 A) = \text{base} + \alpha\Delta$, $P(Y = 1 B) = \text{base} - \alpha\Delta$, with $\text{base}=0.3$, $\Delta = 0.2$. $\alpha = 0$: no difference; $\alpha = 1$: max disparity.
Covariate shift (variance inflation)	$\gamma \in [1, 4]$	Multiply variance of group A ’s feature X_1 by γ , keeping mean fixed.
Asymmetric label noise	$\beta \in [0, 0.3]$	Flip label of group A ’s positive examples with probability β ; group B ’s negative examples with probability $\beta/2$ (asymmetric).

3.6 Analysis

For each scenario, sweep the severity parameter and plot metric values as a function of this parameter. Identify regimes where DP appears to improve while error-based gaps worsen, and relate these patterns to known theoretical results on fairness under shift. Compare observations with qualitative insights from Asiaee (2026), Raab et al. (2021), and follow-up work.

4 Expected Contributions

- A set of **phase diagrams** (heatmaps with shift severity on x-axis, metric value on y-axis, separate panels for each shift type) visualizing how DP and EO diverge under three canonical shift scenarios.
- Identification of “danger zones” — specific shift parameters where DP falsely signals improvement while EO deteriorates (e.g., for asymmetric label noise with $\beta \in [0.15, 0.25]$, DP drops by $> 50\%$ while EO gap increases by 80%).
- A simple **reweighting heuristic** (or diagnostic test) that helps practitioners detect when an observed DP change is likely an artifact.
- Open-source simulation code for reproducibility.

5 What I Expect to See / Why This Is Interesting

I expect to see that:

- Under certain priors or noise patterns, demographic parity can move closer to zero “by accident”, even though the classifier’s behavior is not substantively fairer. This aligns with the invariance result of Asiaee (2026) for error-based metrics but not for acceptance-based metrics.
- Equalized odds will remain stable under prior shift (theoretically expected) but may degrade sharply under label noise (as shown by Raab et al. 2021).

This small simulation study does not compete with full theoretical treatments, but it provides an interpretable “phase diagram” of how fairness metrics behave as data distribution drifts. The goal is to make trade-offs and failure modes more visually and intuitively accessible, and to sanity-check when an observed fairness improvement might be a mirage.

6 Timeline (2 months)

Week	Task
1	Set up synthetic data generator + logistic regression baseline
2	Implement three shift scenarios & sweep parameters
3	Run first batch of experiments (prior shift + covariate shift)
4	Run label noise experiments + compute all metrics
5	Implement reweighting heuristic (importance weighting)
6	Analyze results, generate phase diagrams
7	Draft paper: Intro + Methods
8	Draft Results + Discussion, finalize

References

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